Total Points: [10] [U1HW8 - W10 - Bridge Design]

1. A student designs a bridge with strong truss triangles but decides during construction to just build it with squares because it is faster. What is the most likely outcome?

- The bridge will be much stronger than the planned design.
- It will span a wider gap without using any additional sticks.
- The bridge will suffer from buckling and fail at a much lower weight because it lacks the planned rigidity.
- The efficiency score will automatically reach the maximum possible value.
- The real-world contractor will be praised for their time-saving methods.
- The glue joints will become completely immune to tension forces.

Answer:

2. When a team meticulously calculates forces for a complex bridge design but physically builds a simpler design instead, what essential part of the engineering process is ruined?

- The opportunity to experience compression forces, which only exist in complex designs.
- The ability to predict how the bridge will behave under a heavy load, because the original calculations no longer apply.
- The guarantee that the span will cover 35cm exactly.
- The requirement to use exactly 49 sticks to finish the build.
- The rule that all bridges must have a deck less than 10cm wide.
- The ability to easily paint the popsicle sticks before assembly.

Answer:

3. You are limited to using 49 sticks or less. How does using fewer materials benefit a real-world bridge construction project?

- It eliminates the need for joints in the truss design.
- It makes the bridge significantly heavier and harder to move.
- It increases the time it takes to build the bridge because of careful planning.
- It requires more workers to finish the job safely.
- It ensures that compression forces are maximized in the deck.
- It decreases both the financial cost of materials and the time required to build the structure.

Answer:

4. The 49-stick limit represents a project's real-world budget. Why do structural engineers strive to design a bridge that uses the minimum amount of material necessary?

- Because wood glue only works effectively on small amounts of popsicle sticks.
- Wasting material unnecessarily increases the project's cost, build time, and environmental impact.
- To guarantee the span is longer than 35 centimeters.
- To ensure the bridge is as heavy and difficult to install as possible.
- So the bridge takes longer to build and the contractors get paid for more hours.
- It prevents the need to calculate tension and compression entirely.

Answer:

5. If you build a bridge with 40 sticks instead of 49 sticks, and it holds the exact same amount of weight, how does this affect your bridge's Efficiency Score?

- The score drops because fewer triangles mean weaker joints.
- The score becomes zero because you did not use the maximum allowed material.
- The score decreases because the bridge is physically smaller.
- The score stays exactly the same since it held the same amount of weight.
- The score increases because the bridge itself weighs less, meaning it holds more weight relative to its own mass.
- The score cannot be calculated without knowing the exact width of the bridge.

Answer:

6. Many students created a specific blueprint for their bridge but ended up building something completely different. In the real world, why is it critical for contractors to build exactly what is on the planned design?

- Blueprints are just rough suggestions that contractors can change whenever they want.
- Engineers calculate forces based on that exact design, and changing it unpredictably compromises safety.
- Changing the design during construction is actually encouraged to save money on wood glue.

• It does not matter as long as the bridge looks good when it is finished.

• It ensures the bridge is heavier and therefore much stronger than the blueprint.

• It allows builders to use up extra leftover materials from other jobs freely.

Answer:

7. Imagine your final built bridge is 12cm wide, 30cm long, and uses 55 sticks. Based on the strict project specifications (under 10cm width, 33-35cm long, max 49 sticks), which of the following is true?

• The span length is correct, but the width and material count are wrong.

• It is too narrow, too long, and uses too few sticks.

• It meets all specifications perfectly and is ready for real-world use.

• It is too wide, too short, and uses too much material; a real-world contractor would be fired for this.

• The material count is correct, but the width and length are wrong.

• It is an acceptable prototype because specifications are only recommendations.

Answer:

8. In real-world engineering, what happens if a contractor decides that 33-35cm is too hard to measure, so they build the bridge 38cm long instead?

• The tension forces automatically convert into compression forces, making it stronger.

• The 49-stick limit increases automatically to accommodate the extra length.

• The engineers simply update the blueprint afterwards with no negative consequences.

• The contractor is rewarded for providing extra bridge length at no extra cost.

• The bridge will not fit the pre-built supports, causing massive delays, wasted budget, and safety failures.

• The bridge becomes significantly more efficient because it covers more ground.

Answer:

9. The project specifications state that the bridge must be between 33cm and 35cm

long. Why is it important to follow these exact length dimensions?

- So the bridge can weigh more and hold less weight.
- The specifications are just optional suggestions for beginners.
- To make sure it uses exactly 49 sticks in every situation.
- It allows the bridge to bend sideways easier under compression.
- To ensure the bridge safely spans the physical gap without falling in or wasting expensive material.
- Because longer bridges automatically have higher efficiency scores.

Answer:

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10. The specifications require the bridge width to be under 10cm. What is a real-world consequence if a contractor ignores specific dimensions and just makes them up?

- The bridge's efficiency score will double because it is wider.
- The contractor will receive a bonus for being creative and thinking outside the box.
- Wider bridges always require fewer materials, saving time and money.
- The bridge will automatically be safer because the contractor used their own judgment.
- The bridge might not align correctly with connecting roads, leading to a failed project and legal trouble.
- Tension forces disappear completely when a bridge is wider than 10cm.

Answer:

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